postfix_evaluation.c C Language

```c
#include <stdio.h>
#include <ctype.h>
#include <stdlib.h>

#define MAX 100

int stack[MAX];
int top = -1;

void push(int val) {
    if (top >= MAX - 1) {
        printf("Stack Overflow!\n");
        exit(1);
    }
    stack[++top] = val;
}

int pop() {
    if (top < 0) {
        printf("Stack Underflow!\n");
        exit(1);
    }
    return stack[top--];
}

int evaluatePostfix(char* exp) {
    for (int i = 0; exp[i] != '\0'; i++) {
        if (isdigit(exp[i])) {
            push(exp[i] - '0');
        } else if (exp[i] == ' ' || exp[i] == '\t') {
            continue;
        } else {
            int val2 = pop();
            int val1 = pop();
            switch (exp[i]) {
                case '+': push(val1 + val2); break;
                case '-': push(val1 - val2); break;
                case '*': push(val1 * val2); break;
                case '/': push(val1 / val2); break;
            }
        }
    }
    return pop();
}

int main() {
    char exp[MAX];
    printf("Enter Postfix Expression: ");
    scanf("%s", exp);

    int result = evaluatePostfix(exp);
    printf("Result of Evaluation: %d\n", result);
    return 0;
}
```

Terminal Console Output — Output Verification

```
$ gcc postfix_evaluation.c -o postfix_evaluation
$ ./postfix_evaluation

Enter Postfix Expression: 53+82-*
Result of Evaluation: 48

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Execution Trace / Explanation:
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Input Character | Action                 | Stack Content
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'5'             | Push 5                 | [5]
'3'             | Push 3                 | [5, 3]
'+'             | Pop 3, 5 -> 5 + 3 = 8 | [8]
'8'             | Push 8                 | [8, 8]
'2'             | Push 2                 | [8, 8, 2]
'-'             | Pop 2, 8 -> 8 - 2 = 6 | [8, 6]
'*'             | Pop 6, 8 -> 8 * 6 = 48| [48]
-------------------------------------------------
Final Output    | 48
```